```
clc; clear; close all; % Experiment 1:
Butterworth IIR Filters

Fs = 8000;      % Sampling frequency (Hz)
Ap = 1;         % Passband attenuation (dB)
As = 40;        % Stopband attenuation (dB)

% Low-Pass Filter Specifications
fp_lp = 1000;
fs_lp = 1500;

wp_lp = fp_lp/(Fs/2);
ws_lp = fs_lp/(Fs/2);

[N_lp, Wn_lp] = buttord(wp_lp, ws_lp, Ap, As);
[b_lp, a_lp] = butter(N_lp, Wn_lp, 'low');

% High-Pass Filter Specifications
fp_hp = 2500;
fs_hp = 1800;

wp_hp = fp_hp/(Fs/2);
ws_hp = fs_hp/(Fs/2);

[N_hp, Wn_hp] = buttord(wp_hp, ws_hp, Ap, As);
[b_hp, a_hp] = butter(N_hp, Wn_hp, 'high');

% Display Results
fprintf('BUTTERWORTH IIR FILTER DESIGN\n\n');
```

```
BUTTERWORTH IIR FILTER DESIGN
```

```
fprintf('LOW-PASS FILTER\n');
```

```
LOW-PASS FILTER
```

```
fprintf('Filter Order = %d\n', N_lp);
```

```
Filter Order = 12
```

```
fprintf('Cutoff Frequency = %.2f Hz\n', Wn_lp*Fs/2);
```

```
Cutoff Frequency = 1087.84 Hz
```

```
fprintf('Numerator Coefficients:\n');
```

```
Numerator Coefficients:
```

```
disp(b_lp);
```

```
    0.0000    0.0000    0.0002    0.0006    0.0013    0.0021    0.0024    0.0021    0.0013    0.0006
0.
```

```
fprintf('Denominator Coefficients:\n');
```

```
Denominator Coefficients:
```

```
disp(a_lp);
```

```
    1.0000   -5.4622   14.6771  -25.1616   30.3497  -26.9537   17.9862   -9.0534    3.4018   -0.9283
0.
```

```
fprintf('\nHIGH-PASS FILTER\n');
```

```
HIGH-PASS FILTER
```

```
fprintf('Filter Order = %d\n', N_hp);
```

```
Filter Order = 10
```

```
fprintf('Cutoff Frequency = %.2f Hz\n', Wn_hp*Fs/2);
```

```
Cutoff Frequency = 2379.76 Hz
```

```
fprintf('Numerator Coefficients:\n');
```

```
Numerator Coefficients:
```

```
disp(b_hp);
```

```
    0.0006   -0.0055    0.0248   -0.0662    0.1159   -0.1390    0.1159   -0.0662    0.0248   -0.0055
0.
```

```
fprintf('Denominator Coefficients:\n');
```

```
Denominator Coefficients:
```

```
disp(a_hp);
```

```
    1.0000    1.8915    2.8535    2.6014    1.8704    0.9534    0.3726    0.1030    0.0199    0.0023
0.
```

```
% Frequency Response
[H_lp, f] = freqz(b_lp, a_lp, 2048, Fs);
[H_hp, ~] = freqz(b_hp, a_hp, 2048, Fs);

% Plot Results
figure('Name','Butterworth IIR Filters');

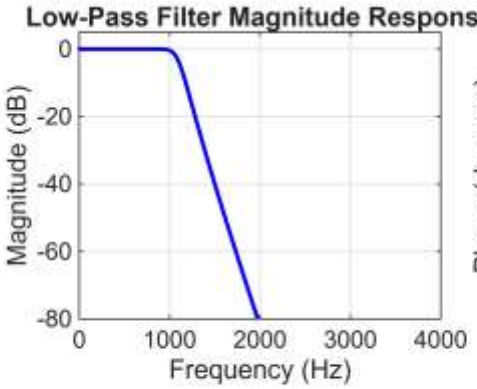
subplot(2,2,1);
plot(f, 20*log10(max(abs(H_lp),1e-8)), 'b', 'LineWidth',1.5);
grid on;
title('Low-Pass Filter Magnitude Response');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-80 5]);
```

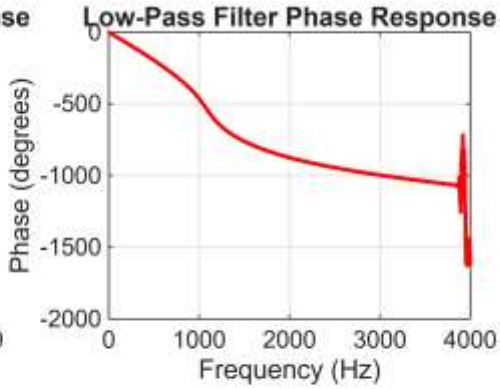
```
xlim([0 Fs/2]);

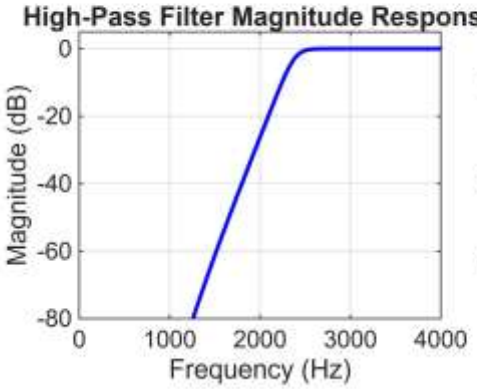
subplot(2,2,2);
plot(f, unwrap(angle(H_lp))*180/pi, 'r', 'LineWidth',1.5);
grid on;
title('Low-Pass Filter Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
xlim([0 Fs/2]);

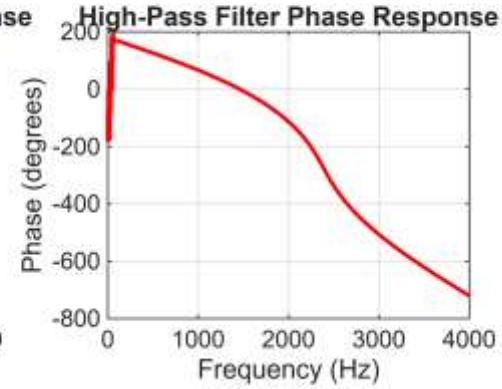
subplot(2,2,3);
plot(f, 20*log10(max(abs(H_hp),1e-8)), 'b', 'LineWidth',1.5);
grid on;
title('High-Pass Filter Magnitude Response');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-80 5]);
xlim([0 Fs/2]);

subplot(2,2,4);
plot(f, unwrap(angle(H_hp))*180/pi, 'r', 'LineWidth',1.5);
grid on;
title('High-Pass Filter Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
xlim([0 Fs/2]);
```


Low-Pass Filter Magnitude Response
Magnitude (dB)
0
-20
-40
-60
-80
0
1000
2000
3000
4000
Frequency (Hz)


Low-Pass Filter Phase Response
Phase (degrees)
0
-500
-1000
-1500
-2000
0
1000
2000
3000
4000
Frequency (Hz)


High-Pass Filter Magnitude Response
Magnitude (dB)
0
-20
-40
-60
-80
0
1000
2000
3000
4000
Frequency (Hz)


High-Pass Filter Phase Response
Phase (degrees)
200
0
-200
-400
-600
-800
0
1000
2000
3000
4000
Frequency (Hz)